

DevOps Home Task

Objective

Design and implement a **complete DevOps pipeline** to deploy a containerized Python web application to AWS using Jenkins and Terraform.

Project Scope

Tech Stack

- **App:** Python (Flask or FastAPI)
 - **OS:** Ubuntu-based EC2 instance (or AMI)
 - **CI/CD:** Jenkins (self-hosted on EC2)
 - **Provisioning:** Terraform
 - **Containerization:** Docker
 - **Cloud:** AWS (EC2 + S3 + IAM + ALB)
-

1 Application Requirements

- A basic Python web API with:
 - 1 GET route (/) returning "Hello, DevOps!"
 - 1 POST route (/echo) returning the input JSON
 - Dockerized with multi-stage Dockerfile
 - Includes `pytest` unit tests
-

2 Infrastructure (Provision via Terraform)

- **VPC** (or default if needed)
 - **1 Jenkins EC2 instance** (Ubuntu)
 - **1 App EC2 instance** (Docker host)
 - **S3 bucket** for artifacts (optional)
 - **Security groups** for Jenkins & app
 - **IAM roles** for Jenkins to access S3 (optional)
-

3 Jenkins CI/CD Pipeline

Set up a **multi-stage Jenkins pipeline** that performs the following:

- ♦ **CI:**
 - Code checkout (GitHub)
 - Lint check (e.g., `flake8`)
 - Run unit tests (`pytest`)

- Build Docker image
- Push image to Docker Hub (or ECR if configured)

♦ **CD:**

- SSH into EC2 App server
- Pull Docker image
- Stop old container, run new container (zero-downtime preferred)
- Health check route after deployment

Use **Jenkinsfile** (declarative pipeline) and store it in your repo root.

4 Infrastructure Automation

Use Terraform to provision:

- All EC2 instances
- S3 bucket (if needed)
- IAM policies & roles
- Output public IPs for Jenkins/App

Use remote state with S3 (optional for bonus)

5 Deliverables

Your GitHub repository must contain:

-  /infra → Terraform code
-  /app → Python source code
-  /jenkins → Jenkins setup scripts (initial setup, plugins, admin user)
-  Jenkinsfile → CI/CD pipeline
-  README.md → Setup instructions & architecture overview

Bonus Points

- Jenkins hosted with a custom domain + HTTPS
- Use ECR instead of Docker Hub
- Use a Docker Compose file for local dev
- Add Slack webhook for deployment success/failure
- Use **systemd** or a **supervisor** for Jenkins persistence
- Enable blue/green deployment (even minimal)

Submission Instructions

- Submit a **link to the public GitHub repo via mail**
- Optional: short Loom video (< 5 min) showing the flow
- Deadline: **29/5 at 20:00**